ZEN CLUB PRESENTS

CODEAMBLE 2026

National Level Hackathon

Patient-Tap

Your medical record, encrypted on the chip you wear.

Team ZEN-T313 · Vishwakarma Institute of Technology

github.com/viralala/Patient-Tap · patient-tap-web.vercel.app

01

Team & Project Overview

Who are you and what are you building?

Team Details

- **Team Name:** Team ZEN-T313
- **Team ID:** ZEN-T313
- **College / Institution:** Vishwakarma Institute of Technology
- **Team Size:** 4 members

Project Overview

Project Title: Patient-Tap

Problem Domain: Select one below

- Agentic & Generative
- Fintech & Cyber
- Agritech & Open
- Smart & Mobility
- **Healthcare & MedTech**
- Blockchain & Web3

02

Problem Statement

What problem are you solving and why does it matter?

The Problem

When disaster strikes or the network goes down, first responders treat patients blind — with no access to their allergies, medications or DNR wishes. The result is preventable drug reactions and duplicated care as a patient is handed between responders, ambulances and field hospitals. Today's medical-ID apps live in the cloud or a locked phone, so they fail at the exact moment they are needed most.

Scale / Impact

≈1.3M people are harmed by medication errors every year, at an estimated $42B global cost (WHO). In emergencies, missing history is a leading — and avoidable — risk.

Who is Affected

Unconscious or non-verbal emergency patients, people with allergies or chronic conditions, the elderly — and the paramedics forced to treat them with no record at all.

Why Unsolved

Cloud apps need a network disasters take down. Phone Medical IDs need a charged, unlocked device. Engraved bracelets hold two static lines — no updates, no treatment log.

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03

Proposed Solution

What are you building and how does it work?

Our Solution

Patient-Tap stores the patient's full medical record — encrypted with AES-256-GCM — directly on the NFC chip they wear (an NTAG215 card or wristband, ≤504 bytes). Any phone reads or writes it in under a second with zero network, and responders log the treatment they give straight back onto the tag.

Encrypted on the Chip

The record lives on the tag itself, sealed with AES-256-GCM. No cloud, no account — it works even with no signal, and a tampered tag fails the auth check.

Offline Conflict Checker

Before a responder administers a drug, Patient-Tap checks it against the patient's allergies and recent doses on-device, and blocks dangerous conflicts behind an explicit override.

Patient + Responder App

One app, two modes: patients build and write their profile; responders scan, treat and log back to the tag. Native Android NFC, with a web build for demos.

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04 Innovation & Differentiators

What makes your solution stand out from the rest?

★ Key Differentiators

What makes your solution unique? How is it different from existing solutions? List 2–3 key differentiators below:

→ **Offline by design:** the record reads in under a second on the chip — no network, no account — exactly where cloud and phone IDs fail.

→ **Built-in safety check:** an on-device conflict checker blocks allergy and dose clashes before administration — a life-saver no medical-ID rival offers.

→ **Travels with the patient:** responders write treatment back onto the tag as a portable chain-of-custody; each NTAG215 tag costs only a few rupees.

⚖ Vs Existing Solutions

Feature	Existing Solutions	Our Solution
Works fully offline	✗	✓
Affordability	⚠	✓
Responder write-back	✗	✓
Drug-conflict check	✗	✓

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05

Impact & Benefits

Who benefits and how will you measure success?

Expected Impact

Faster, safer emergency care for anyone who cannot speak for themselves — with fewer preventable drug reactions and no duplicated treatment. Because it needs no network, the same record works in disaster zones, rural areas and ambulances alike. Success = decrypt-to-display in under one second, offline, with every field a responder needs.

Target Users

Emergency & chronic patients, the elderly, and first responders / paramedics.

Scale / Reach

Any NFC-capable phone — billions of devices; cent-cost tags deployable even in low-infrastructure regions.

Economic Impact

Cuts costly adverse-drug-event care; tags cost a few rupees; tiered plans (Free / ₹999 Premium / ₹1999 B2B).

Social Impact

Advances SDG 3 (Good Health) and personal health-data sovereignty — equitable emergency care where networks don't reach.

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06

Market Research

Validate demand and identify opportunities

Competitive Landscape

The emergency medical-ID market is real but network-dependent. The leading options — and where each one stops short:

→ MedicAlert — engraved bracelet + cloud profile: static, subscription-based, and needs a network to fetch the full record.

→ Apple Health Medical ID — phone lock-screen info: useless if the phone is dead, absent or locked, and no responder can write back to it.

→ Govt digital health IDs (e.g. ABHA) — cloud EHR: online-only, not emergency-first, with no offline tag or in-field treatment logging.

Gap Analysis

Validate demand and identify opportunities

Gap Analysis

Where existing emergency medical-ID solutions fall short — and how Patient-Tap closes each gap:

Parameter	Existing Solutions	Patient-Tap
Works offline	Cloud-dependent — fails in a blackout	**Full record on the chip, zero network**
Data location	Cloud, or a locked / dead phone	**Encrypted on the NFC tag you wear**
Treatment log	None — not updatable in the field	**Responders write back; chain-of-custody**
Drug safety	Manual guesswork under pressure	**On-device conflict check with override**
Cost & access	Subscriptions; no field hardware	**Cent-cost NTAG215 + freemium app**

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